

What we're doing

Can whole-contact-map clustering identify coherent rollout basins well enough that partial maps from cluster medoids improve oracle best-of-100 contact accuracy?

Why

#326's pairwise contact clusters failed to improve held-out oracle accuracy. Whole-map clustering is a stronger test: it defines modes using complete predicted structures and branches from spatially distributed contacts copied from actual medoid maps.

The primary objective is oracle best-of-100, not consensus. The 16-protein development gate requires +0.005 versus both iid100 and a same-size random-map seed control before any of the 81 confirmation proteins are scored.

Results so far

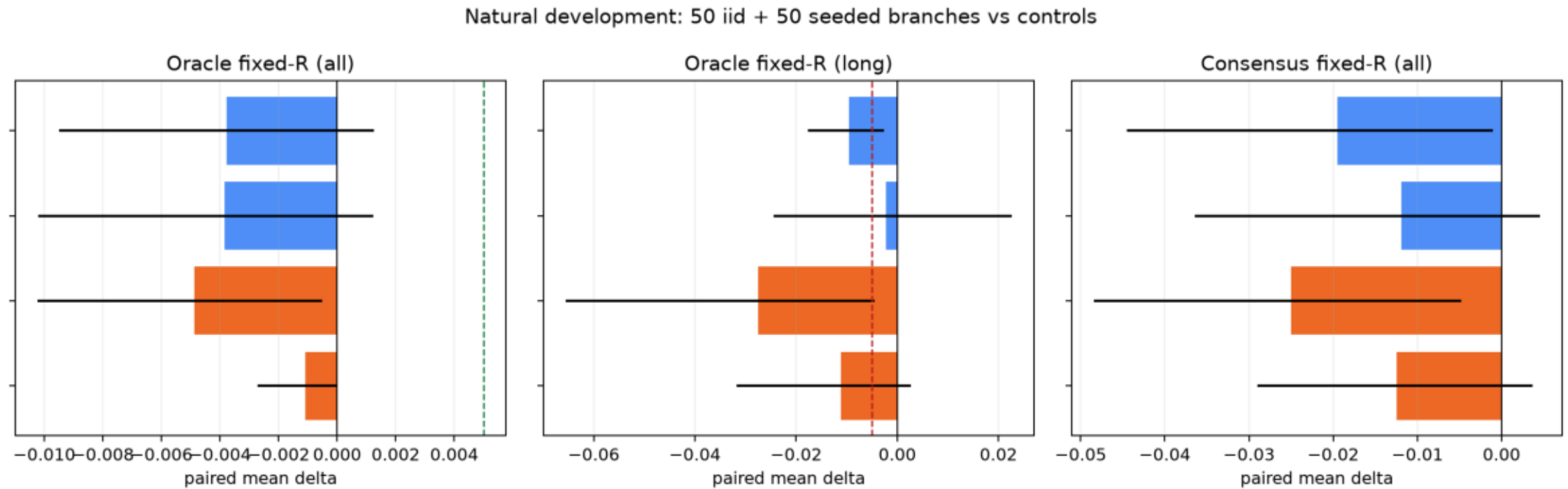
The development gate failed, so the 81 confirmation proteins remain unread.

- 8-contact medoid oracle: -0.0038 vs iid100 and -0.0039 vs random-map control. -
16-contact medoid oracle: -0.0049 vs iid100 and -0.0011 vs random-map control. - Long-
range oracle: -0.0096 and -0.0274 vs iid100. - Consensus: -0.0195 and -0.0250 vs iid100.

Medoid seeds were much less accurate than random-map seeds (18% vs 31% at k=8; 17% vs 29% at k=16). The clusters capture coherent model errors, not useful under-sampled conformations. Fold-switch movement was no better than random seeding. Do not adopt this brancher.

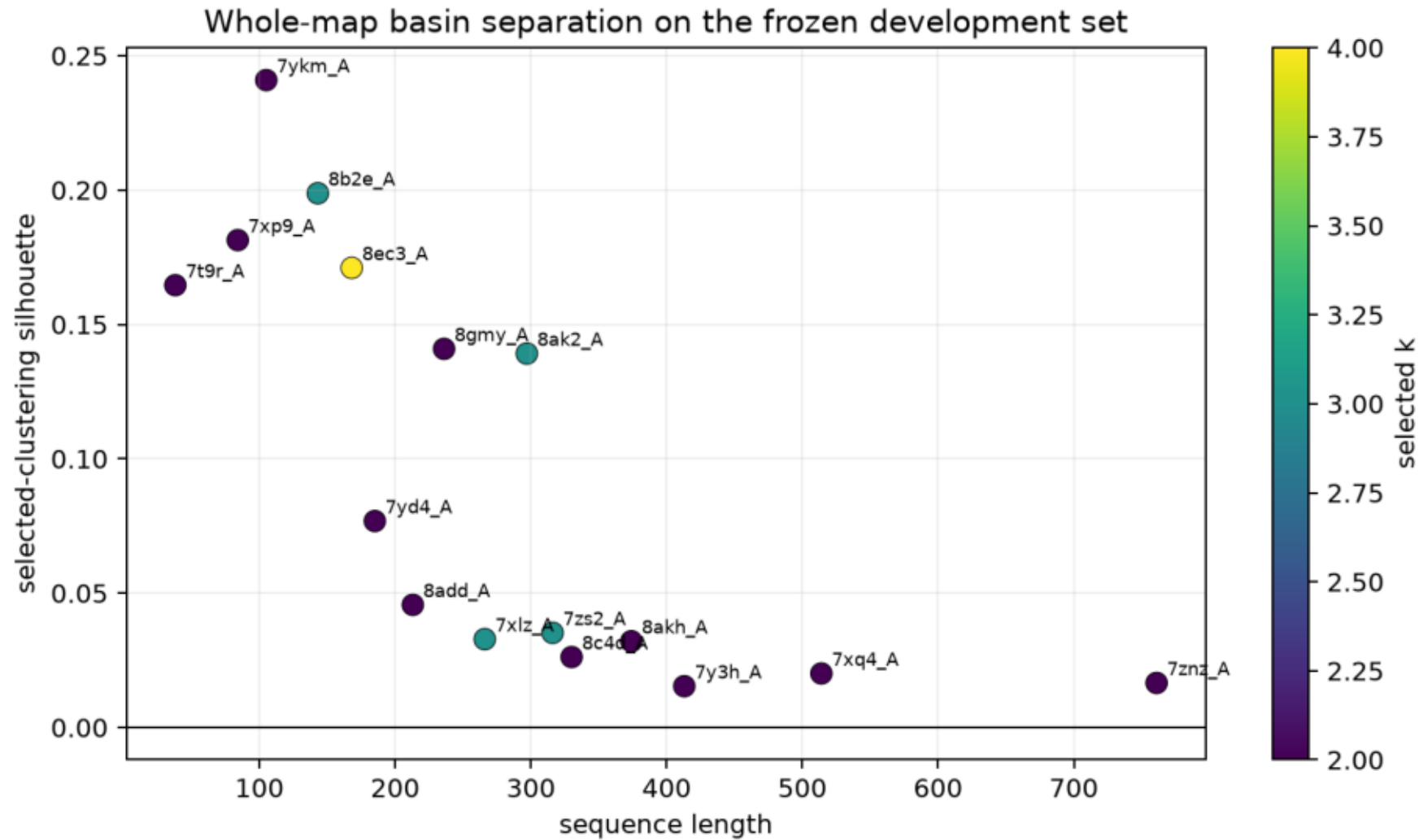
01_dev_paired_deltas.png

Protein-paired development deltas with 95% bootstrap intervals. Green/red dashed lines mark the preregistered all/long gate thresholds.



02_cluster_diagnostics.png

Silhouette of the deterministic k-medoids choice for each protein; the two seed sizes share the same clustering.



03_foldswitch_modes.png

Number of frozen fold-switch pairs for which each 100-map pool reaches each mode or both modes.

